

Comparing K-Anonymity and Differential Privacy in Practice: An Evaluation Across Three Public Datasets

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Abstract

Data anonymization is important for enabling safe sharing of personal data under legal frameworks such as GDPR and CCPA. Two major techniques, k-anonymity and differential privacy, are widely used to justify that anonymized data can be freely shared, but the question of which technique actually performs better in practice remains questioned. The paper implements both techniques on three real, publicly available datasets of varying size and domain. The Adult Census Income dataset, the Heart Disease (Cleveland) dataset, and the Diabetes Health Indicators dataset. K-anonymity is implemented using the Mondrian multidimensional partitioning algorithm tested at k values of 2, 5, 10, 25, and 50. Differential privacy is implemented using the Laplace mechanism through IBM's Diffprivlib library at epsilon values of 0.1, 0.5, 1.0, 2.0, 5.0, and 10.0. Both techniques are evaluated on a unified framework: classifier accuracy (utility) and exact matching reidentification rate (privacy), enabling direct comparison on the same axes. Results show that differential privacy preserves significantly more utility than k-anonymity on two of three datasets, with accuracy advantages of 5.74 percentage points on Adult Census Income and 5.39 percentage points on Heart Disease, while both techniques achieve equivalent near zero reidentification rates across all parameter settings. On the large Diabetes dataset, both techniques perform pretty much the same. These results suggest that differential privacy is generally the better approach, and also give useful advice on how to decide which of the two methods to use.

1. INTRODUCTION

Data anonymization is one of the most important and widely misunderstood problems in modern computer science. As personal data is collected at an extreme scale through smartphones, medical systems, and online platforms, the ability to share that data safely for research has become very important. Under frameworks like the General Data Protection Regulation (GDPR) and the California Consumer Privacy Act (CCPA), correctly anonymized data can be stored, shared, and used for any purpose without requiring individual consent. The problem is that the techniques used to justify that claim are increasingly shown to be weaker than assumed.

Two of the most leading techniques are k-anonymity and differential privacy. K-anonymity, introduced by Sweeney (2002), guarantees that every record in a released dataset is identical from at least k-1 others across a set of quasi-identifying attributes fields like age, sex, and zip code that do not directly identify someone but can be combined to do so. Differential privacy, formalized by Dwork (2006), provides a mathematically provable guarantee: the output of a computation does not reveal whether any individual's data was included, regardless of what background knowledge an adversary already has.

While differential privacy is widely described as theoretically better, the practical question of whether it performs better than k-anonymity on real datasets remains unknown. Gadotti et al. (2024) identify this gap in their review, arguing that evidence connecting formal guarantees to real world performance is lacking. El Emam et al. (2009) studied k-anonymity but never got to compare it to differential privacy on the same data. This project addresses that gap directly.

Both techniques are implemented in Python and applied to three publicly available datasets of varying size and data types. They are evaluated on a unified framework, classifier accuracy as the utility measure and exact matching reidentification rate as the privacy measure, placing both techniques on the same axes to enable direct head to head comparison. The main research question is, does differential privacy preserve more data utility than k-anonymity while providing equivalent or better privacy protection in practice, and does that hold across different types of datasets?

The paper is organized as follows... Section 2 reviews five related papers. Section 3 describes the datasets, implementations, and evaluation design in enough detail to reproduce the work. Section 4 presents and compares results. Section 5 discusses findings and limitations. Section 6 concludes with future work.

2. BACKGROUND

2.1 Anonymization Landscape (Gadotti et al., 2024)

Gadotti et al. (2024) provide a comprehensive survey of modern anonymization in "Anonymization: The Imperfect Science of Using Data While Preserving Privacy." The authors connect legal definitions, mathematical theory, and real world examples in this paper, arguing that k-anonymity cannot scale to modern data. A formal proof they cite shows that even weak versions of k-anonymity severely destroy utility when there are as few as 10 to 15 identifying attributes. The paper presents differential privacy as the strongest formal framework available. It takes a bunch of scattered information and pulls it together into one clear argument. What it does not do is run new experiments or provide guidance on which technique to use in practice, the exact gap this project fills.

2.2 K-Anonymity Model (Sweeney, 2002)

Sweeney (2002) introduced k-anonymity as a formal model for protecting privacy in released datasets. The core guarantee is that any record in the published data matches at least k-1 other records on all quasi-identifying attributes, requiring an adversary to guess among at least k candidates. Sweeney demonstrates reidentification of attacks against voter registration and medical records, establishing foundational vocabulary for quasi-identifiers, sensitive attributes, and generalization hierarchies. The limitation is that k-anonymity does not address adversaries with background knowledge, a gap later addressed by l-diversity and t-closeness and does not compare against probabilistic techniques like differential privacy.

2.3 K-Anonymity in Practice (El Emam et al., 2009)

El Emam et al. (2009) looks at k-anonymity distinguishes two threat scenarios... the prosecutor scenario (where the adversary knows the target is in the dataset) and the journalist scenario (where the adversary does not). They show k-anonymity is overly conservative for the journalist scenario, destroying more information than necessary. Three modifications are evaluated to preserve more utility at equivalent risk. What the paper does well is this threat model distinction, which most work barely looks over. What it does not do is compare k-anonymity to differential privacy at all, leaving the most obvious follow up question unanswered, which is exactly the question this paper answers.

2.4 Differential Privacy (Dwork, 2006)

Dwork (2006) introduced differential privacy as a formal guarantee: a mechanism M satisfies ϵ -differential privacy if for any two datasets D and D' differing by one record, and any output S , $P[M(D)=S] \leq e^{\epsilon} * P[M(D')=S]$. This means the result looks almost the same whether one person's data is in it or not, even if someone has extra outside info, making it stronger than k-anonymity. The Laplace mechanism provides a concrete implementation by adding noise scaled to query sensitivity divided by ϵ . The paper establishes the theoretical main standard but provides no comparison to existing methods like k-anonymity on the same datasets.

2.5 Beyond K-Anonymity (Li et al., 2007)

Li, Li, and Venkatasubramanian (2007) introduce t-closeness to address a weakness in l-diversity: even l-diverse datasets can leak information if the distribution of sensitive values within a partition differs from the global distribution. T-closeness requires that within partition distributions match the global distribution within a threshold measured by earth mover's distance. This matters because k-anonymity needs more and more complicated fixes to deal with its weaknesses, while differential privacy offers one clear, consistent way to protect privacy. It motivates using differential privacy as the primary comparison point rather than any of the k-anonymity variants.

3. APPROACH

3.1 Datasets

Three publicly available datasets were chosen to test whether results generalize across different sizes, data types, and

attribute types. All preprocessing was done in Python using pandas. Each dataset was split 80/20 into a training set (anonymized) and a held out test set (used for utility evaluation and reidentification attacks) using graded sampling with `random_state = 42` to ensure reproducibility.

Adult Census Income (UCI Machine Learning Repository): 32,561 records, 15 attributes. After dropping rows with missing values and removing `fnlwgt` and `native_country` as non informative for anonymization purposes, 30,162 records and 13 attributes were left. Categorical columns (`workclass`, `education`, `marital_status`, `occupation`, `relationship`, `race`, `sex`) were encoded to integer codes using pandas `Categorical`. The binary target variable indicates `income > $50K`. Quasi-identifiers (QIs): `age`, `sex`, `race`, `education_num`, `marital_status`, `hours_per_week`. Train: 24,129 records. Test: 6,033 records.

Heart Disease, Cleveland (UCI): 303 records, 14 attributes. After removing 6 records with missing values in the `ca` and `thal` columns, 297 records remained. Target binarized: 0 = no disease (original value 0), 1 = disease present (original values 1-4). QIs: `age`, `sex`, `cp` (chest pain type), `trestbps` (resting blood pressure), `chol` (cholesterol), `thalach` (max heart rate). Train: 237 records. Test: 60 records.

Diabetes Health Indicators (Kaggle, BRFSS 2015 survey): 253,680 records, 22 attributes, no missing values. Target: `Diabetes_binary` (renamed to 'target'). All columns converted to integer. QIs: `Age`, `Sex`, `BMI`, `Education`, `Income`, `GenHlth`, `PhysHlth`, `MentHlth`. Train: 202,944 records. Test: 50,736 records. This dataset is included specifically to test whether results are held at large scale.

3.2 K-Anonymity: Mondrian Algorithm

K-anonymity was implemented from scratch in Python using pandas, based on the Mondrian multidimensional partitioning algorithm (LeFevre et al., 2006). The full implementation is available in `kanonymity.py` file. The algorithm works as follows:

- (1) Start with all training records as a single partition.
- (2) For the current partition, compute the normalized range of each QI column: $(\max - \min) / (\text{global max} - \text{global min} + 1e-10)$. Select the column with the largest normalized range as the split dimension.
- (3) Split on the median of that column in the current partition, creating left ($\leq$ median) and right ($>$ median) sub partitions.
- (4) If either sub partition has fewer than k records, keep the current partition unsplit. Otherwise, recurse on both sub partitions.
- (5) After all of the partitions are finalized, replace every QI value in each partition with the mean of that column within that partition. All records in the same partition become identical on QIs.

Information loss is computed as: $IL = (1 / (N * |QI|)) * \sum_{\text{partitions}} \sum_{QI} (\text{partition_range} / \text{global_range}) * \text{partition_size}$, where N is the total number of records. IL ranges from 0 (no generalization) to 1 (all values collapsed to a single number).

Parameter sweep: $k \in \{2, 5, 10, 25, 50\}$. Each k value produces a separate anonymized CSV saved to `results/kanonymity/`.

3.3 Differential Privacy: Laplace Mechanism

Differential privacy was implemented using IBM's Diffprivlib library (Holohan et al., 2019) (version 0.6.6), applying the Laplace mechanism independently to each QI column. The full implementation is in the differentialprivacy.py file. The process for each QI column col is:

(1) Compute global sensitivity as $\text{sensitivity} = \text{col.max()} - \text{col.min()}$. This approximates the worst case change in the column's value caused by adding or removing one record. (2) Instantiate a Laplace mechanism: `dp.mechanisms.Laplace(epsilon= $\epsilon$ , sensitivity=sensitivity)`. (3) For each value v in the column, draw a noisy value: $v' = v + \text{Laplace}(0, \text{sensitivity}/\epsilon)$. (4) Clip v' to $[\text{col.min()}, \text{col.max()}]$ to prevent values outside the original range.

Average noise is reported as: $\text{avg_noise} = (1/|QI|) * \sum QI \text{mean}(|\text{noisy} - \text{original}|) / \text{column_range}$. This normalizes noise by column range, making it comparable across attributes with different scales.

Parameter sweep: $\epsilon \in \{0.1, 0.5, 1.0, 2.0, 5.0, 10.0\}$. Each epsilon produces a separate anonymized CSV saved to results/diffprivlib/. Noise is random: results will vary slightly between runs due to random Laplace sampling.

3.4 Unified Evaluation Framework

The evaluation design addresses the key weakness identified in the original proposal. Without a common measurement scale, you cannot directly answer which technique is better. All metrics are computed identically for both techniques on all three datasets using evaluate.py.

Utility - Classifier Accuracy: A logistic regression model (scikit-learn LogisticRegression, `max_iter=1000`, `random_state=42`) is trained on the anonymized training set. Features are standardized using StandardScaler fit on the anonymized training data. The model is evaluated on the original (un anonymized) held out test set. This checks how useful the anonymized data still is for making predictions on a real task.

Accuracy drop = $\text{baseline_accuracy} - \text{anonymized_accuracy}$, where lower drop is better.

Privacy - Exact Matching Reidentification Attack: For each record in the held out test set, its QI values are rounded to 2 decimal places and matched against the anonymized training set. A record is reidentified if exactly one record in the anonymized training set shares all its QI values, a unique match means an adversary who has both datasets can link the test record to a specific individual with certainty. $\text{Reidentification rate} = \text{reidentified_count} / \text{total_test_records}$.

Baseline accuracy is computed by training logistic regression on the original (un anonymized) training data and evaluating the original test set. This serves as the upper bound for utility.

Both techniques are plotted on the same axes (reidentification rate vs. classifier accuracy) in Figure 1, placing every k setting and every ϵ setting as a single point on the same graph. The technique whose points cluster higher (more utility) at

equivalent or lower reidentification rate is better. A summary (Figure 5) reports the accuracy difference DP - k -anonymity per dataset.

4. EVALUATION

4.1 Baselines

Logistic regression on original unanonymized data: Adult = 0.8178, Heart = 0.8333, Diabetes = 0.8621. These are the upper bounds on utility for any anonymization technique.

4.2 K-Anonymity Results

Table 1 summarizes k -anonymity information loss across datasets. Adult and Heart show high information loss (0.65 and 0.70 respectively) because the Mondrian algorithm must generalize heavily to create equal sized partitions in a relatively small number of dimensions with skewed distributions. Diabetes shows lower information loss (0.39–0.42) because its large size allows many tight partitions.

Table 1. K-Anonymity Information Loss by Dataset and k

$k=2$: Adult 0.653, Heart 0.702, Diabetes 0.394 | $k=5$: 0.653, 0.702, 0.398 | $k=10$: 0.653, 0.702, 0.403 | $k=25$: 0.653, 0.740, 0.412 | $k=50$: 0.658, 0.740, 0.423

On Adult Census Income, k -anonymity averaged 0.7411 accuracy across all k values (drop: 7.67 pp). Accuracy was nearly flat from $k=2$ (0.7436) to $k=10$ (0.7436), then declined slightly to 0.7373 at $k=50$. This plateau occurs because Mondrian partitioning on this dataset reaches a structural floor early. The dataset naturally clusters into a small number of QI-homogeneous groups regardless of k in the lower range.

On Heart Disease, k -anonymity averaged 0.7933 accuracy (drop: 4.00 pp). Results were flat at 0.7667 for $k=2$ through $k=10$, then jumped to 0.8333 at $k=25$ and $k=50$. The jump reflects Mondrian merging partitions when the small dataset (237 records) cannot handle detailed splits at higher k .

On Diabetes, k -anonymity averaged 0.8609 accuracy (drop: 0.12 pp), declining smoothly from 0.8616 at $k=2$ to 0.8603 at $k=50$. The large dataset allows for detailed partitioning at all k values tested, resulting in minimal utility loss.

4.3 Differential Privacy Results

Differential privacy noise decreases cleanly as epsilon increases across all three datasets. At $\epsilon=0.1$ (strongest privacy), average normalized noise was approximately 0.48–0.49 across all datasets. At $\epsilon=10.0$ (weakest privacy), noise dropped to 0.07–0.08.

On Adult, DP averaged 0.7985 accuracy across all ϵ settings (drop: 1.93 pp). Accuracy improved from 0.7948 at $\epsilon=0.1$ to 0.8129 at $\epsilon=10.0$, showing that things steadily improve as the noise goes down.

On Heart, DP averaged 0.8472 accuracy, exceeding the baseline of 0.8333 at five of six ϵ settings (drops were negative: -0.0167). This is discussed in Section 5.

On Diabetes, DP averaged 0.8601 accuracy (drop: 0.21 pp), remaining nearly flat across all ϵ values (range: 0.8598 to

0.8610). Because the dataset is so large, the extra noise doesn't really affect the results of logistic regression much.

4.4 Privacy: Reidentification Results

Figure 4 shows the reidentification heatmap. K-anonymity achieved exactly 0.0000 reidentification across all datasets and all k settings. Differential privacy achieved 0.0000 for Adult and Heart at all epsilon values, and 0.0001 for Diabetes at $\epsilon=0.1$ and $\epsilon=0.5$ (approximately 5 records out of 50,736 test records reidentified). Both techniques are effectively equivalent on privacy under the exact matching attack.

This equivalence is a key finding because neither technique fails on privacy in this experimental setting and utility becomes the only differentiating dimension between them.

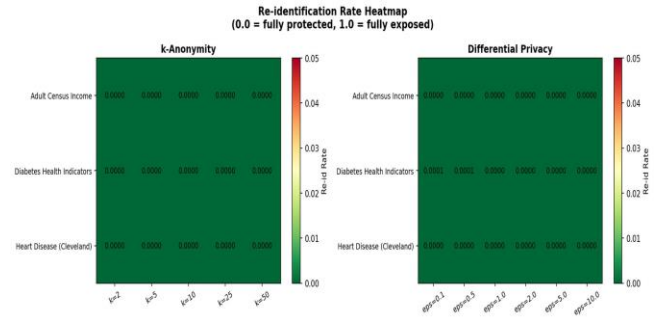


Figure 4. Reidentification rate heatmap for both techniques across all datasets and parameter settings. All cells are near zero, confirming equivalent privacy protection.

4.5 Direct Utility Comparison

Figure 1 plots all parameter settings for both techniques on the same Privacy vs. Utility axes. On Adult and Heart, the red DP diamonds cluster consistently higher than the blue k-anonymity circles at equivalent reidentification rates near zero, showing DP is strictly better on utility. On Diabetes, the clusters overlap closely.

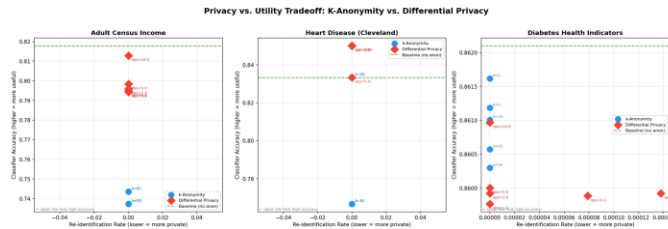


Figure 1. Privacy vs. Utility tradeoff for both techniques across all three datasets. Each point represents one parameter setting. Higher accuracy = more useful. Lower reidentification rate = more private.

Figure 2 shows accuracy drop as a function of parameter strength. The k-anonymity line (blue) consistently sits higher than the DP line (red) on Adult and Heart, meaning k-anonymity loses more utility at every parameter setting. On Diabetes, both lines are nearly indistinguishable.

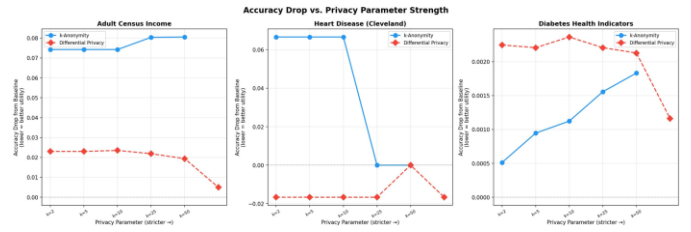


Figure 2. Accuracy drop from baseline as a function of privacy parameter strength. Lower is better (less utility loss). K-anonymity parameters increase left to right ($k=2$ to $k=50$); DP parameters are overlaid on the same x-axis positions.

Figure 3 shows average accuracy across all parameter settings per technique per dataset. DP (red) outperforms k-anonymity (blue) on Adult (0.799 vs. 0.741) and Heart (0.847 vs. 0.793). On Diabetes both are within 0.001 of each other.

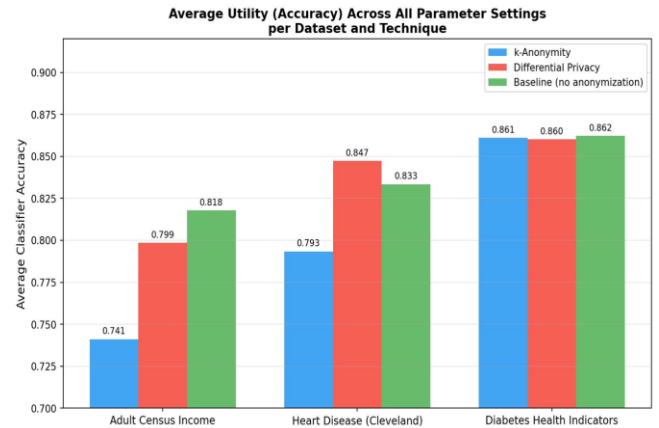


Figure 3. Average classifier accuracy across all parameter settings per dataset and technique, compared to the no-anonymization baseline (green).

Figure 5 summarizes the dominance result directly. Differential privacy wins on Adult (+5.74 pp) and Heart (+5.39 pp). K-anonymity wins on Diabetes by a margin of 0.09 pp, negligible in practice.

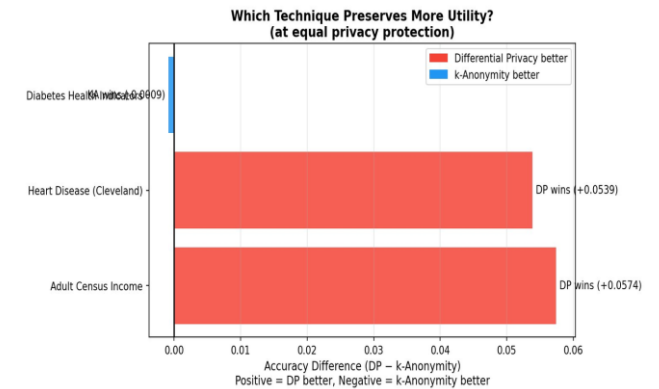


Figure 5. Accuracy difference (DP - k-anonymity) per dataset. Positive = DP better, negative = k-anonymity better.

5. DISCUSSION

5.1 Why Differential Privacy Performs Better

K-anonymity generalizes by replacing all QI values within a partition with that partition's mean. Every record in the same partition becomes identical on quasi-identifiers, collapsing all within partition variation. A logistic regression classifier

relies on variation between records to learn decision boundaries. Variation that is destroyed by generalization cannot be recovered. This is why k-anonymity's accuracy floor is so much lower than DP's.

Differential privacy adds Laplace noise to each record individually. The global statistical distribution of the dataset is largely preserved, means and variances shift only slightly, while individual records become harder to identify. The logistic regression classifier can still learn meaningful decision boundaries from data that is noisy but statistically representative of the original. This is why DP accuracy losses are small, and why they decrease predictably as epsilon increases.

This pattern is consistent with the theoretical prediction of Gadotti et al. (2024). K-anonymity destroys information structurally and irreversibly, while differential privacy reduces it in a controlled way that you can adjust with settings.

5.2 The Diabetes Exception

With 202,944 training records and QIs that are mostly categorical or ordinal with narrow value ranges (e.g., Age has 13 levels, Sex is binary), Mondrian produces many small, tight partitions that generalize minimally information loss of 0.39–0.42, compared to 0.65 for Adult and 0.70 for Heart. When partitions are small and tight, the mean replacement generalization barely changes the data. At the same time, DP's noise at low epsilon values is large enough to reduce accuracy by 0.21 pp on average, slightly more than k-anonymity's 0.12 pp. The practical takeaway is that k-anonymity can work well on very large datasets when the identifying fields have few unique values and don't vary much.

5.3 Heart Disease: DP Exceeds Baseline

Differential privacy achieved higher accuracy than the unanonymized baseline on Heart Disease at five of six epsilon settings. With only 237 training records, the logistic regression model trained on the original data may overfit slightly. The Laplace noise added by DP acts as a form of regularization, smoothing the decision boundary. This happens because the dataset is small, not because of differential privacy in general, so don't take it as DP actually making the data better.

5.4 Limitations

The exact matching attack is the simplest possible reidentification method because both techniques reduce QI precision (through generalization or noise), exact matches become rare regardless of the privacy parameter. More advanced attacks, like guessing based on probabilities, isolating individuals, or combining with other datasets, would be more effective at telling the difference between these methods and would likely show bigger gaps between them. The near zero reidentification rates observed here should not be seen as absolute privacy guarantees.

The Mondrian implementation generalizes using partition means rather than range intervals, which is a simplification.

Basic k-anonymity methods group data into broader ranges using predefined intervals. The Laplace mechanism is applied independently per attribute, ignoring correlations between QI columns. Both are reasonable approximations that are sufficient for comparison but not the most optimized versions of each technique.

6. CONCLUSION AND FUTURE WORK

The project implemented k-anonymity (Mondrian algorithm) and differential privacy (Laplace mechanism) in Python and applied both to three publicly available datasets: Adult Census Income (30,162 records), Heart Disease — Cleveland (297 records), and Diabetes Health Indicators (253,680 records). Both techniques were evaluated on a unified framework, logistic regression classifier accuracy as the utility measure and exact matching reidentification rate as the privacy measure, placing every parameter setting from both techniques on the same axes to enable direct comparison.

Differential privacy preserves significantly more utility than k-anonymity at equivalent privacy protection on two of three datasets: +5.74 percentage points on Adult Census Income and +5.39 percentage points on Heart Disease. On Diabetes, both techniques perform equivalently (+0.09 pp in favor of k-anonymity, not practically meaningful). Both techniques achieve near zero reidentification rates under exact matching attacks across all parameter settings, making utility the decisive differentiating dimension.

These findings provide support for the theoretical claims of Gadotti et al. (2024) and offer actionable guidance. Practitioners should prefer differential privacy when utility preservation matters. K-anonymity can still perform well on very large datasets when the identifying fields are few in numbers and don't have many unique values.

There are several directions for future work. First, replacing exact matching with probabilistic or background knowledge attacks would better reflect realistic adversary capabilities and likely show stronger differentiation between the techniques. Second, implementing interval based generalization for k-anonymity (rather than mean replacement) and composition aware DP mechanisms would test fairer, more optimized versions of each approach. Third, extending the evaluation to l-diversity or t-closeness would place both techniques in the broader anonymization landscape established by Li et al. (2007). Fourth, testing on higher dimensional datasets with 50 or more quasi-identifiers would directly probe the scalability failure of k-anonymity predicted by Gadotti et al. (2024).

REFERENCES

Dwork, C. (2006). Differential privacy. In Automata, Languages and Programming: 33rd International Colloquium, ICALP 2006, Lecture Notes in Computer Science (Vol. 4052, pp. 1–12). Springer. https://doi.org/10.1007/11787006_1

El Emam, K., Jabbouri, S., Sams, S., Drouet, Y., & Power, M. (2009). Protecting privacy using k-anonymity. *Journal of the American Medical Informatics Association*, 16(5), 627–637. <https://pmc.ncbi.nlm.nih.gov/articles/PMC2528029/>

Gadotti, A., Rocher, L., Houssiau, F., Cretu, A., & de Montjoye, Y. (2024). Anonymization: The imperfect science of using data while preserving privacy. *Science Advances*, 10(29). <https://doi.org/10.1126/sciadv.adn7053>

Li, N., Li, T., & Venkatasubramanian, S. (2007). t-Closeness: Privacy beyond k-anonymity and l-diversity. In *Proceedings of the IEEE 23rd International Conference on Data Engineering (ICDE)* (pp. 106–115). IEEE.

Sweeney, L. (2002). k-Anonymity: A model for protecting privacy. *International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems*, 10(5), 557–570.

LeFevre, K., DeWitt, D. J., & Ramakrishnan, R. (2006). Mondrian multidimensional k-anonymity. In *Proceedings of the 22nd International Conference on Data Engineering (ICDE)* (p. 25). IEEE.

Holohan, N., Braghin, S., Mac Aonghusa, P., & Levacher, K. (2019). Diffprivlib: The IBM differential privacy library. *arXiv preprint arXiv:1907.02444*. <https://arxiv.org/abs/1907.02444>